



CENTRAL UNIVERSITY OF SOUTH BIHAR

Department of Computer Science

Class Test-I, 25th Sept. 2025

Subject: Computer Networks

Course Code: CSC81DC00304 Semester: Ist

Time: 1 Hour

M. M.: 10

Session: 2025-2027

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1. What do you mean by OSI? Explain the 7 OSI layers in order. (4 Marks)
 2. Why HTTP is a Stateless Protocol? (2 Marks)
 3. Write Short notes on (any two): (4 Marks)
(A) Network Edge & Network Core (B) FTP (C) Packet Switching



CENTRAL UNIVERSITY OF SOUTH BIHAR

Department of Computer Science

Class Test-II, 20th Nov. 2025

Subject: Computer Networks

Course Code: CSC81DC00304 Semester: Ist

Time: 1 Hour

M. M.: 10

Session: 2025-2026

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1. What do you mean by multiplexing techniques? Explain. (4 Marks)
 2. Define network topology. Explain the basic types of topology. (2 Marks)
 3. Write Short notes on (any two): (4 Marks)
(A) IEEE 802.11 (B) UA and MTA (C) Ethernet



CENTRAL UNIVERSITY OF SOUTH BIHAR
Department of Computer Science
End Semester Examination: Dec 2025

Programme: M.Sc.(CS)
Course Title: Computer Networks
Maximum Points: 70
Enrollment Number:

Course code: CSC81DC00304
Duration: 3:00 Hrs
Semester – 1st

Instruction:

1. Attempt questions in all the Sections as directed.

Section - A

Answer all questions from below, each carrying 1 point.

(10 X 1 = 10)

1. Define Computer Network.
2. What are the two types of line configuration?
3. Why are protocols needed?
4. What is an Internet?
5. What is the Network Edge and the Network Core?
6. What is DNS?
7. List out the name of Network Connecting devices.
8. What are UA and MTA in SMTP?
9. Why HTTP is a *Stateless Protocol*?
10. Draw a neat diagram of a Communication Model.

Section - B

Answer any 5 questions below, each carrying 6 points.

(5 X 6 = 30)

1. What is Network Topology? Explain different types of topologies in Network. [3+3]
2. The XYZ Corporation has a fully connected mesh network consisting of five devices. Calculate the total number of cable links needed.
3. Compare HTTP and FTP.
4. What are the three traditional switching methods? What are the two approaches to packet-switching? [3+3]
5. What is BOOTP? Why DHCP is required? [3+3]

6. Five channels, each with a 200 kHz bandwidth, are to be multiplexed together. What is the minimum bandwidth of the link if there is a need for a guard band of 25 kHz between the channels to prevent interference? Also draw a neat diagram.

Section - C

Answer any 2 questions below, each carrying 15 Points. (2 X 15 = 30)

1. Explain OSI network architecture and explain the functionalities of every layer in detail?
[15 Points]
2. Describe the following (any two):
[7.5+ 7.5 Points]
 - I. CSMA/CD and Ethernet
 - II. Explain TCP/IP model with a neat diagram
 - III. IEEE 802.11
3. What is Multiplexing? Describe the goals of multiplexing. Explain the three multiplexing techniques in detail.
[15]